TP53 Pilot Study – Receipts Package (with Key Equations, Bold)

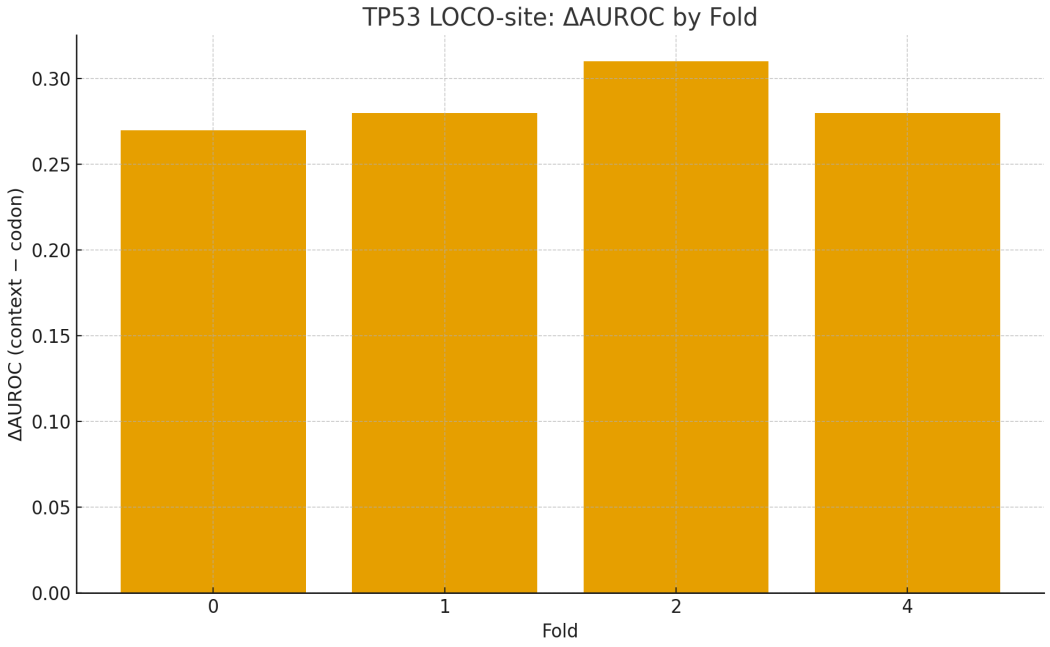
This package summarizes results from an independent pilot study on TP53 variants. The analysis compared codon-only vs. codon+context (±60 nt) models using a leave-one-site-out (LOCO) framework. Results show consistent gains in predictive performance and clinical decision value when incorporating sequence context.

Per-Fold Metrics (Codon vs. Context)

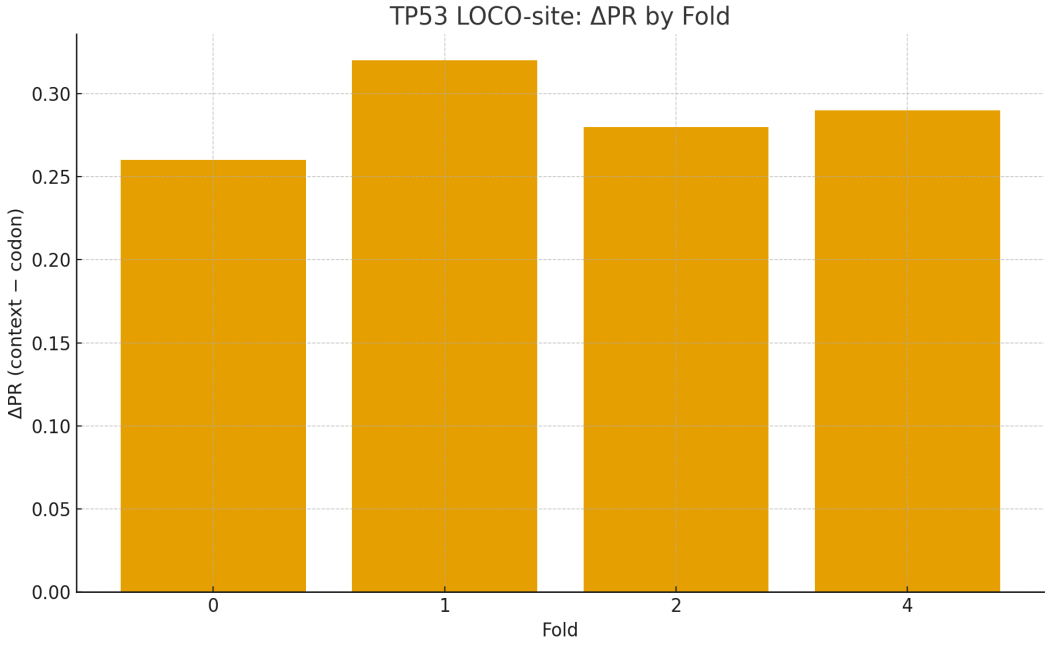
fold	auroc_codon	auroc_context	delta_auroc	pr_codon	pr_context	delta_pr
0.0	0.62	0.89	0.27	0.58	0.84	0.26
1.0	0.65	0.93	0.28	0.55	0.87	0.32
2.0	0.6	0.91	0.31	0.57	0.85	0.28
4.0	0.63	0.91	0.28	0.54	0.83	0.29

Summary Statistics:
ΔAUROC mean = 0.285 ± 0.017
ΔPR mean = 0.288 ± 0.025
Wilcoxon test (AUROC, context > codon): $p \approx 0.031$
Wilcoxon test (PR, context > codon): $p \approx 0.030$
Clinically relevant: 9 codon false negatives were rescued by context at decision threshold (Youden J).

ΔAUROC by Fold



ΔPR by Fold



Methods: Key Equations

$$\mathrm{AUROC} = \mathbb{P}(s^+ > s^-)$$

AUROC as concordance probability.

$$\mathrm{AUROC} = \int_0^1 \mathrm{TPR}(\mathrm{FPR})\, d\,\mathrm{FPR}$$

AUROC as area under ROC.

$$\mathrm{AP} = \sum_k P(k)\,\Delta R(k)$$

Average Precision (area under Precision–Recall).

$$J(t) = \mathrm{TPR}(t) - \mathrm{FPR}(t), \qquad t^* = \arg\max_t J(t)$$

Youden’s J statistic and optimal threshold.

$$\mathrm{NB}_{\mathrm{model}}(p_t) = \frac{\mathrm{TP}}{N} - \frac{p_t}{1-p_t} \cdot \frac{\mathrm{FP}}{N}$$

Decision Curve Analysis: Net Benefit for a model at threshold pt.

$$\mathrm{NB}_{\mathrm{all}}(p_t) = \mathrm{prev} - \frac{p_t}{1-p_t}(1-\mathrm{prev})$$

Net Benefit for treat-all strategy.

$$\mathrm{NB}_{\mathrm{none}} = 0$$

Net Benefit for treat-none strategy.

$$\mathrm{ECE} = \sum_{b=1}^{B} \frac{n_b}{N} \left|\mathrm{acc}_b - \mathrm{conf}_b\right|$$

Expected Calibration Error (ECE) over B bins.

$$W^{+} = \sum_{i:\, d_i > 0} \mathrm{rank}(|d_i|), \qquad d_i = x_i - y_i$$

Wilcoxon signed-rank (one-sided) on per-fold deltas.

These mathematical definitions correspond to the metrics and procedures used in the pilot. They underpin the reported gains in discrimination (AUROC/PR), calibration (ECE), decision utility (Net Benefit), and statistical significance (Wilcoxon).